Support Vector Machine (SVM)

Key Points

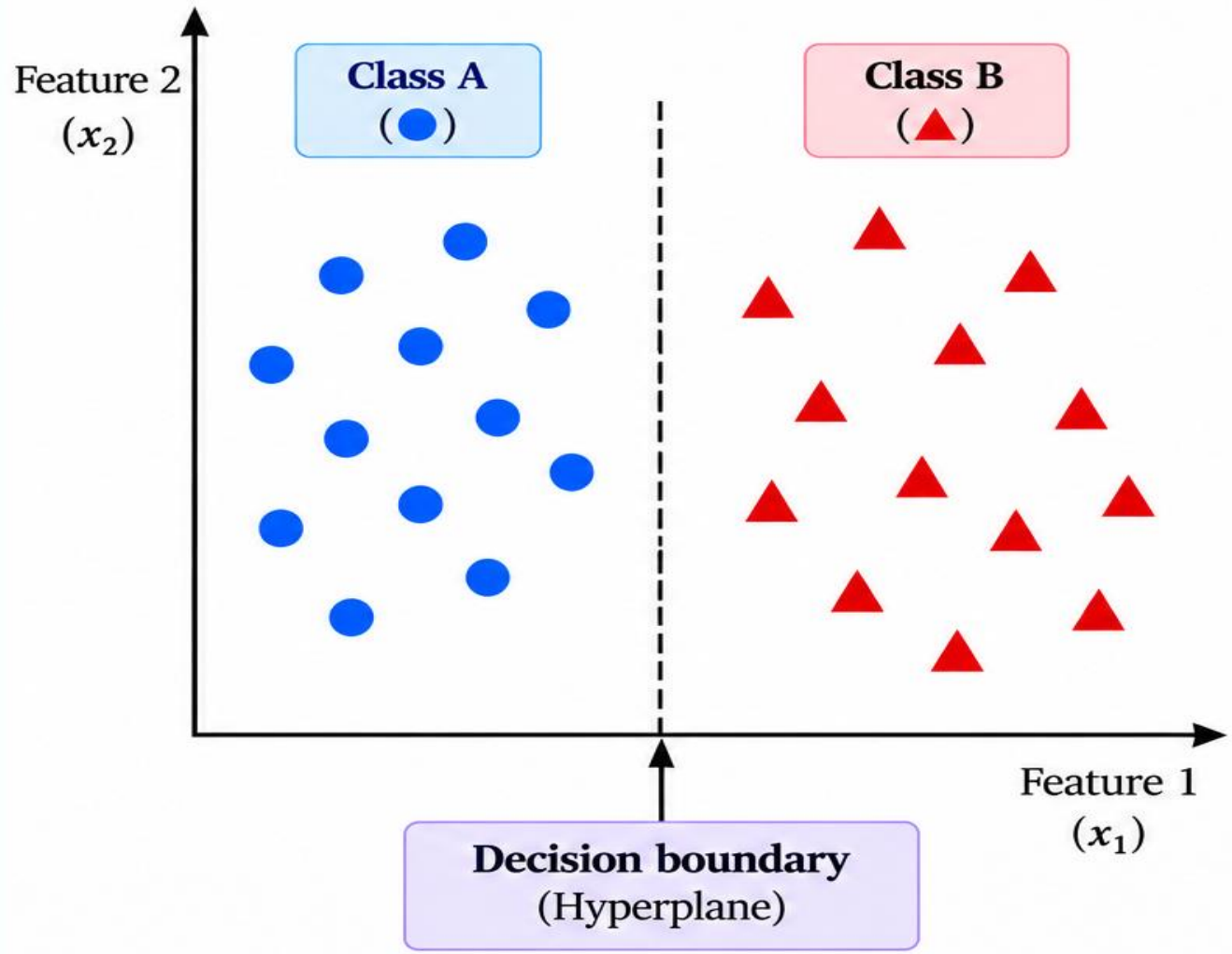
- Support Vector Machine (SVM) is a supervised machine learning algorithm mainly used for **classification** and also for **regression**.
- SVM finds a suitable decision boundary, called a **hyperplane**, that separates data points belonging to different classes.
- In a two-dimensional space, the hyperplane is a **straight line**. In higher dimensions, it becomes a **hyperplane**.
- SVM tries to find a boundary (hyperplane) that separates the two classes with the maximum possible margin.

Two Classes

Class A ●

Class B ▲

Two Classes Separated by a Hyperplane



Hyperplanes

Definition

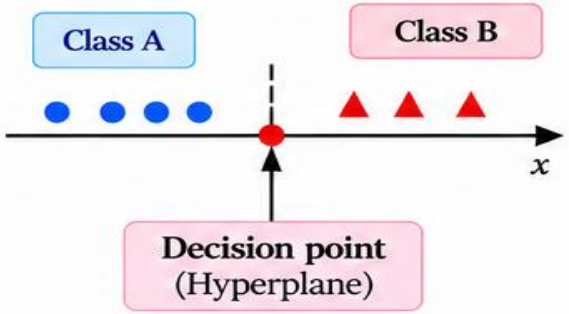
- A **hyperplane** is a boundary or surface used to separate data points belonging to different classes.
- In SVM, the hyperplane acts as the **decision boundary** between classes.

Simple Definition | A **hyperplane** is a decision boundary that separates different classes of data.

Hyperplane in Different Dimensions

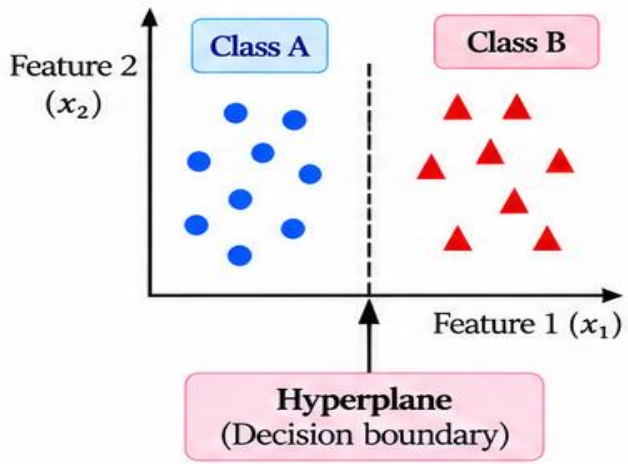
1. One-dimensional data

The hyperplane is a **point**.



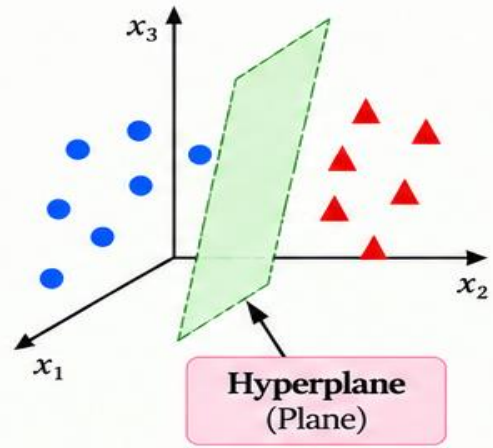
2. Two-dimensional data

The hyperplane is a **straight line**.



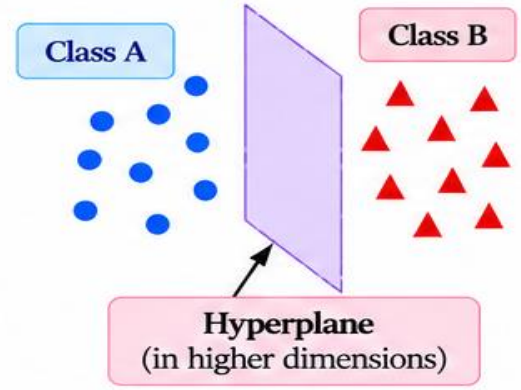
3. Three-dimensional data

The hyperplane is a **plane**.



4. Higher-dimensional data

For more than three features, the separating boundary is called a hyperplane.



Key Point

A hyperplane divides the feature space into two regions, with each region belonging to a different class.

3 Equation of a Hyperplane

- The general equation of an SVM hyperplane is:

$$w^T x + b = 0$$

- For two features:

$$w_1 x_1 + w_2 x_2 + b = 0$$

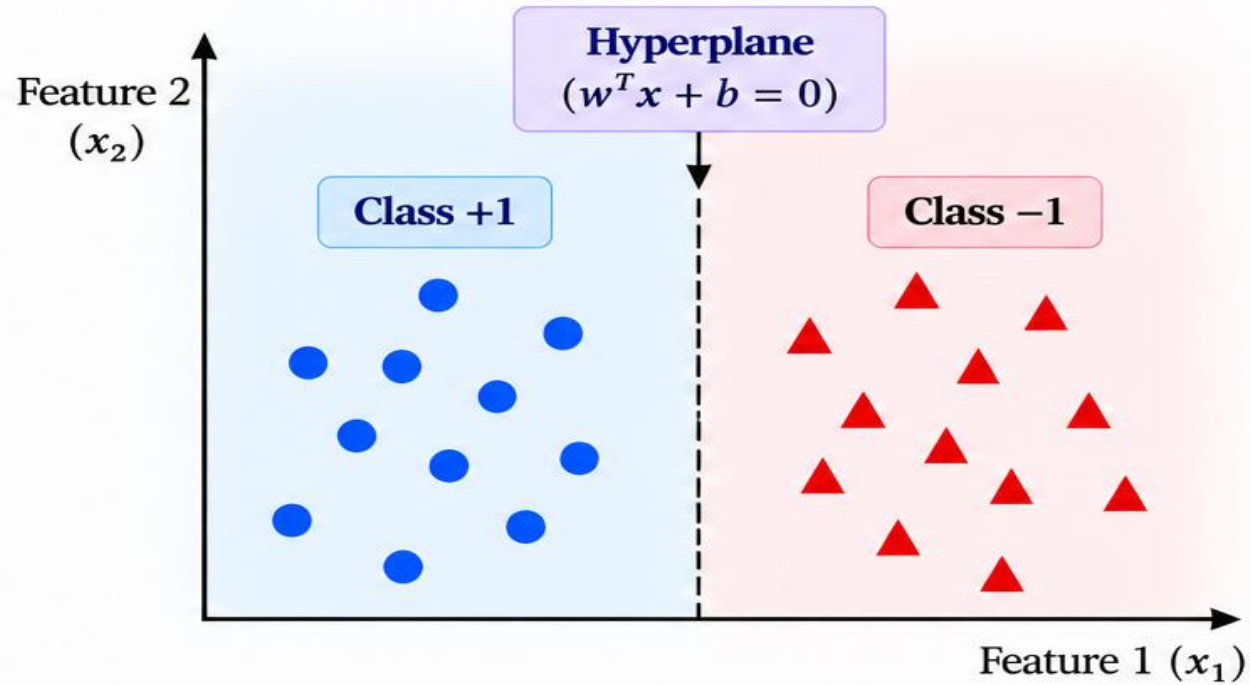
where:

- x_1, x_2 = input features
- w_1, w_2 = weights
- b = bias/intercept
- w = weight vector
- x = input vector

4 How a Hyperplane Classifies Data

- The hyperplane divides the data space into two regions.
- For the hyperplane $w^T x + b = 0$, the classification can be represented as:

$$\begin{cases} +1, & w^T x + b > 0 \\ -1, & w^T x + b < 0 \end{cases}$$



A point on one side of the hyperplane belongs to one class, while a point on the other side belongs to the other class.

Example

1 Given Features

- x_1 = Study Hours
- x_2 = Attendance

2 Hyperplane

$$2x_1 + x_2 - 10 = 0$$

3 For a Student

$$x_1 = 4, \quad x_2 = 5$$

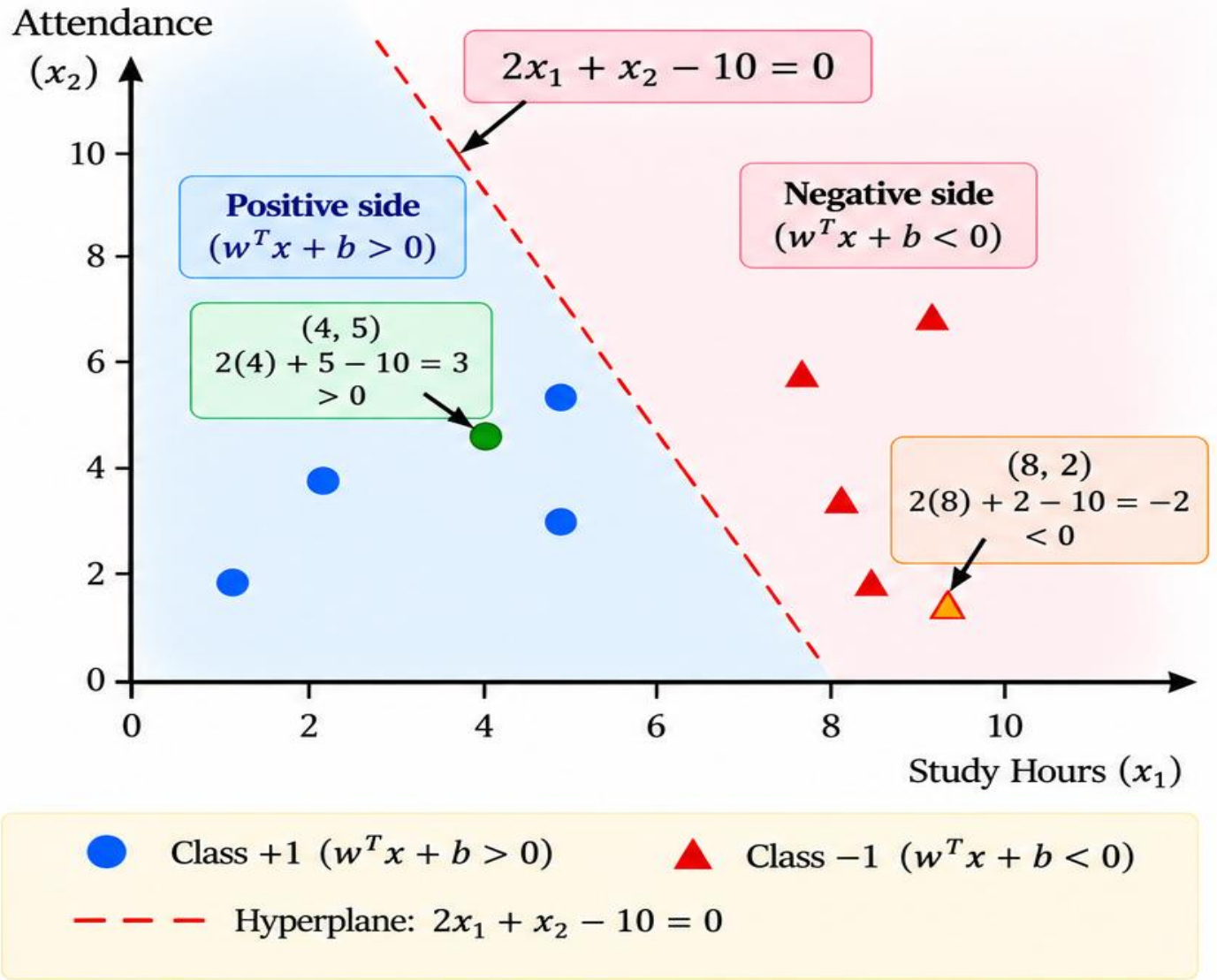
4 Substitute the Values

$$2(4) + 5 - 10 = 3$$

5 Result

- $3 > 0$
- The point lies on the **positive side** of the hyperplane.
- If another point gives a negative value, it lies on the **opposite side**.

Graphical Representation



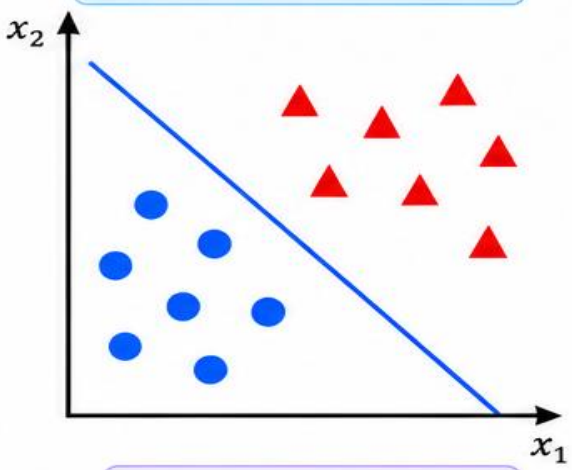
Orientation of a Hyperplane

- The **weight vector** w determines the **orientation** of the hyperplane, while the **bias** b determines its **position**.
- For example: $w_1x_1 + w_2x_2 + b = 0$
- Changing w_1 or w_2 changes the **direction/orientation** of the boundary.
- Changing b **shifts** the boundary.

1 Changing w_1 or w_2 (Orientation)

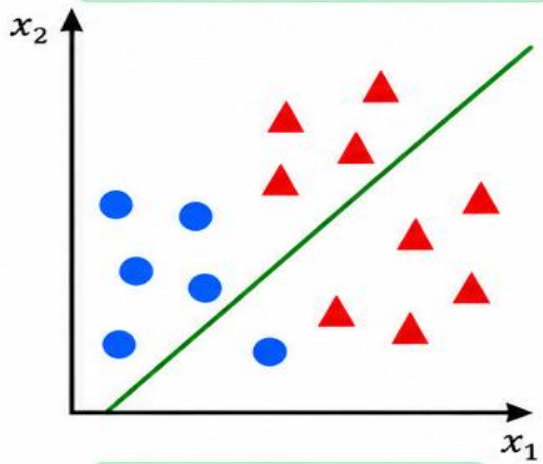
Changing the values of w_1 or w_2 changes the direction (orientation) of the hyperplane.

$w_1 = 1,\ w_2 = 1,\ b = 0$



Hyperplane:
$x_1 + x_2 = 0$

$w_1 = 1,\ w_2 = -1,\ b = 0$

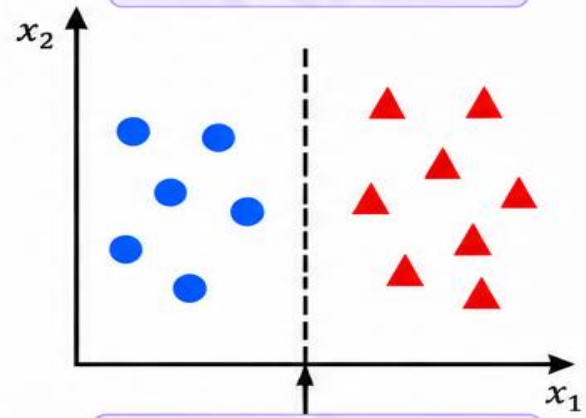


Hyperplane:
$x_1 - x_2 = 0$

2 Changing b (Position)

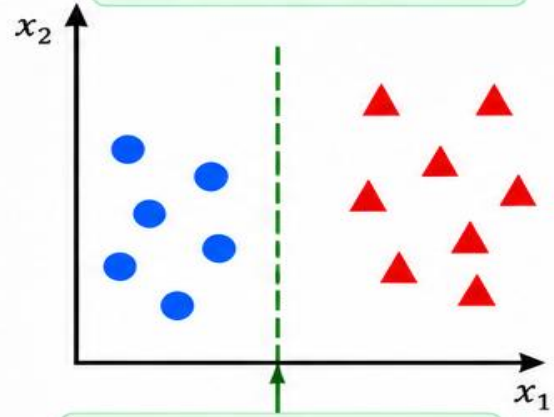
Changing the value of b **shifts** the hyperplane without changing its orientation.

$b = 0$
(original position)



Hyperplane:
$w_1x_1 + w_2x_2 + 0 = 0$
(e.g., $x_1 = 0$)

$b = -2$
(shifted position)

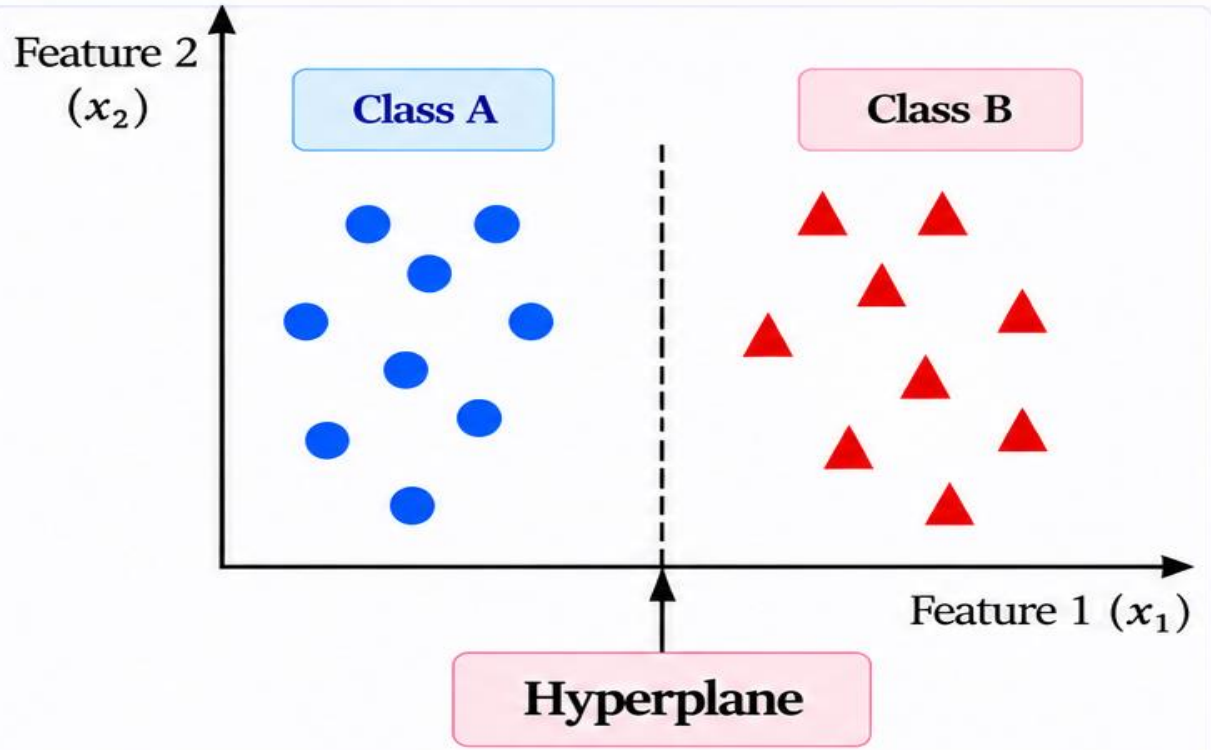


Hyperplane:
$w_1x_1 + w_2x_2 - 2 = 0$
(shifted to the right)

Maximum Margin Hyperplanes in Linearly Separable Data

1 Linearly Separable Data

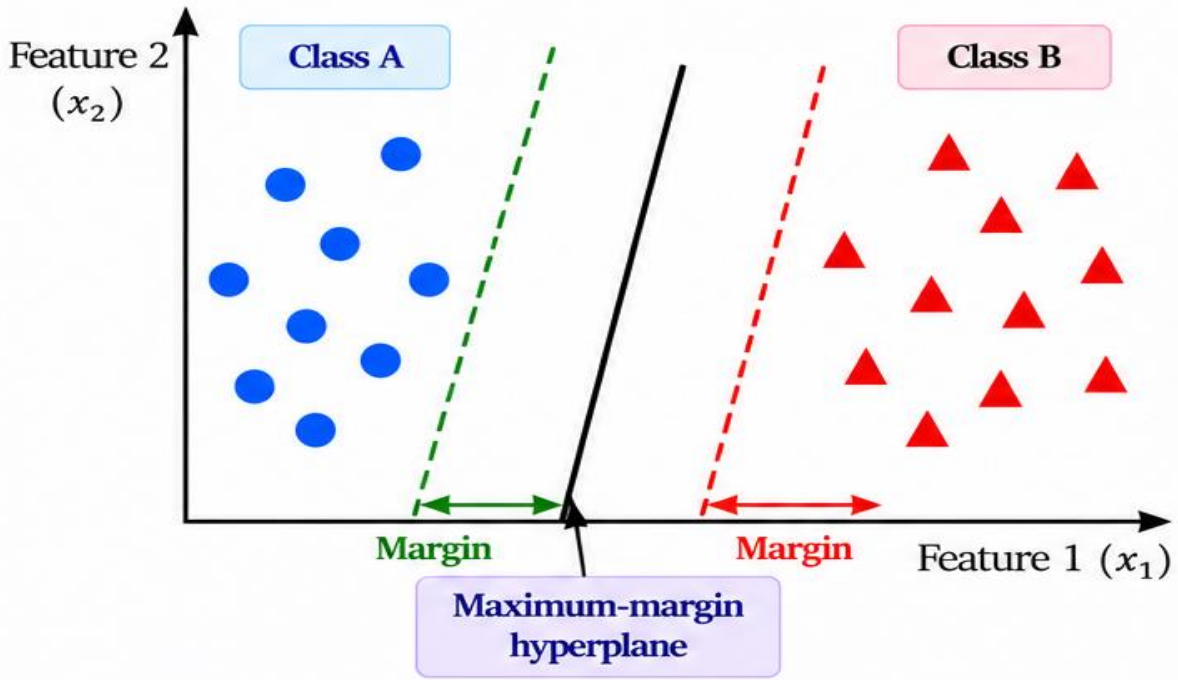
- Data is called **linearly separable** when a straight line can completely separate two classes without any data points being on the wrong side.
- For two-dimensional data, the two classes can be separated using a **single straight line**.



Simple Definition | Linearly separable data is data that can be divided into different classes using a single straight line or hyperplane.

2 Maximum Margin Hyperplane

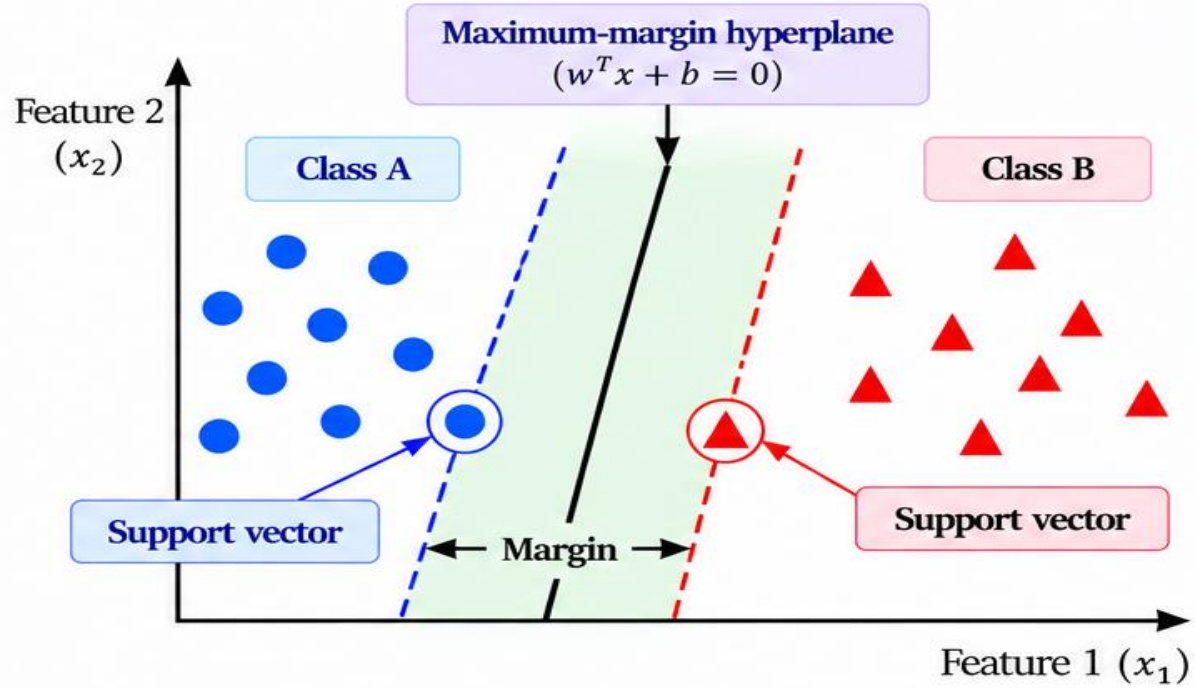
- There can be several hyperplanes that separate linearly separable data.
- SVM selects the hyperplane that gives the **maximum margin**.
- The central line is the **maximum-margin hyperplane**.



Class A
Class B
Maximum-margin hyperplane $(w^T x + b = 0)$
Margin boundary $(w^T x + b = -1)$
Margin boundary $(w^T x + b = +1)$

3 Margin and Support Vectors

- The **margin** is the distance between the decision hyperplane and the closest data points from each class.
- SVM tries to make this distance as large as possible.
- The data points closest to the decision boundary are called **support vectors**.



Class A
Class B
Maximum-margin hyperplane $(w^T x + b = 0)$
Margin boundary $(w^T x + b = -1)$
Margin boundary $(w^T x + b = +1)$
Support vector (closest point to hyperplane)

4 Three Important Lines in SVM

For a linearly separable SVM, we consider **three parallel lines**:

1 Decision hyperplane

$$w^T x + b = 0$$

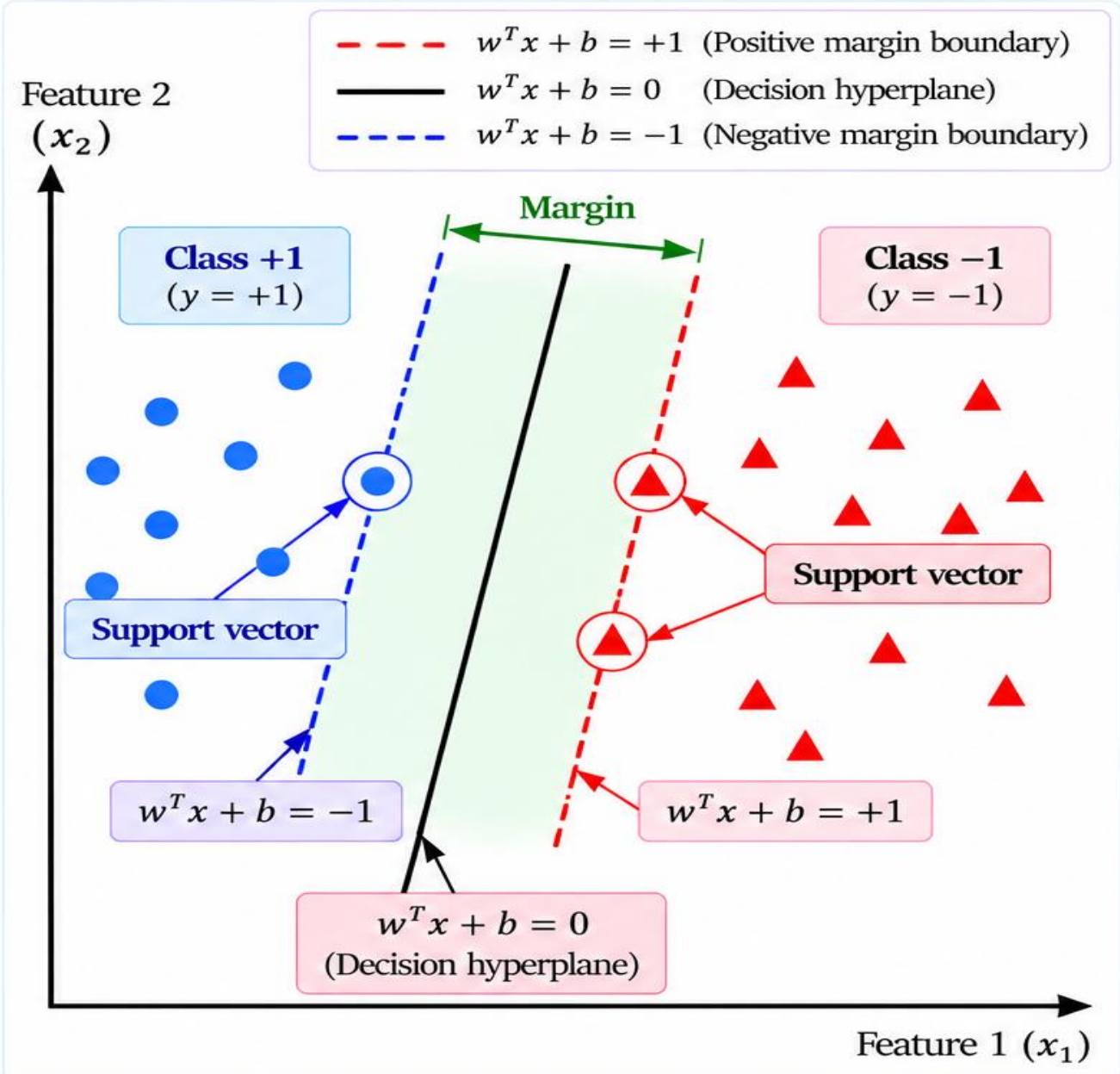
2 Positive margin boundary

$$w^T x + b = +1$$

3 Negative margin boundary

$$w^T x + b = -1$$

- These three lines are parallel to each other.
- The middle line ($w^T x + b = 0$) is the **decision hyperplane**.
- The points that lie on the two margin boundaries ($w^T x + b = +1$ and $w^T x + b = -1$) are the **support vectors**.



5 Margin Formula

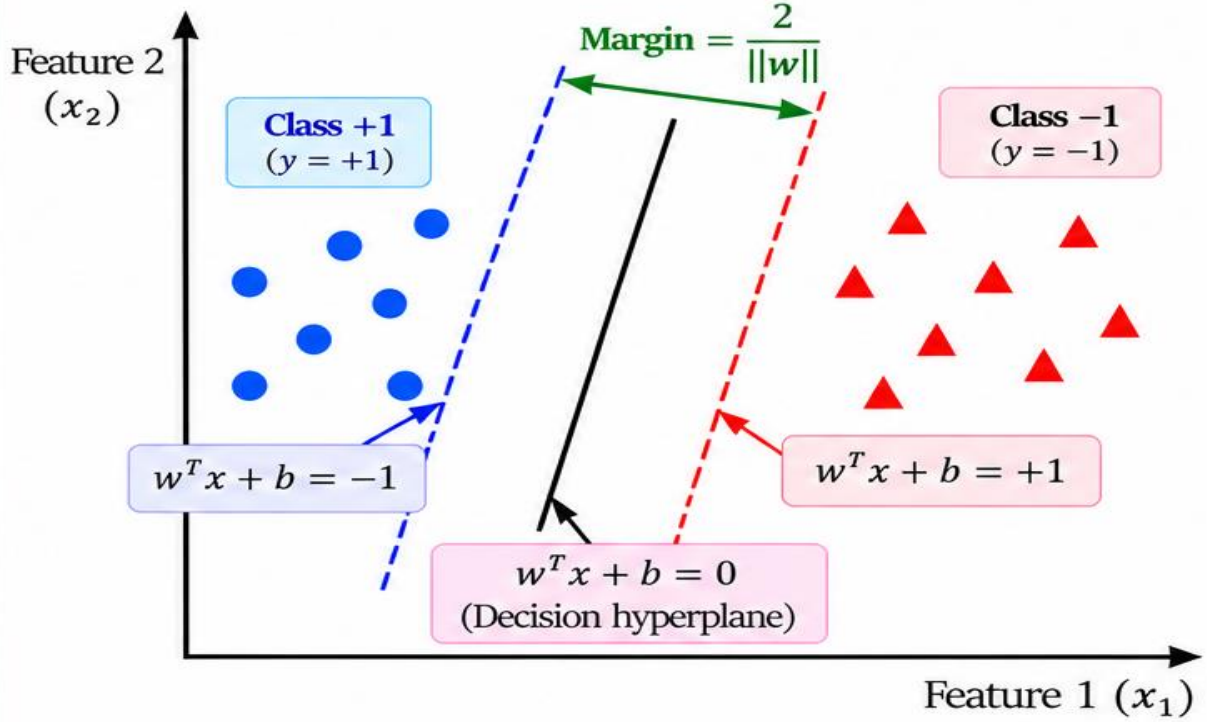
- For the hyperplane: $w^T x + b = 0$
- The distance between the two margin boundaries is:

$$\text{Margin} = \frac{2}{||w||}$$

- Therefore, to obtain the maximum margin, SVM minimizes:

$$\frac{1}{2}||w||^2$$

subject to the classification constraints.



6 Classification Constraint

- For linearly separable training data, each training example has a class label:

$$y_i \in \{-1, +1\}$$

- The SVM requires:

$$y_i(w^T x_i + b) \geq 1$$

- This ensures that the training points are correctly classified and lie outside or on their corresponding margin boundary.

Meaning of the Constraint

For class +1 $(y_i = +1)$	→ $w^T x_i + b \geq 1$ (lies on or beyond the positive margin)
For class −1 $(y_i = -1)$	→ $w^T x_i + b \leq -1$ (lies on or beyond the negative margin)

Geometric Interpretation

- All training points must lie outside or on the margin boundaries $w^T x + b = +1$ and $w^T x + b = -1$.
- Points that lie exactly on these boundaries are the **support vectors**.
- The constraint helps to correctly classify the training data with the maximum possible margin.

5 Margin Formula

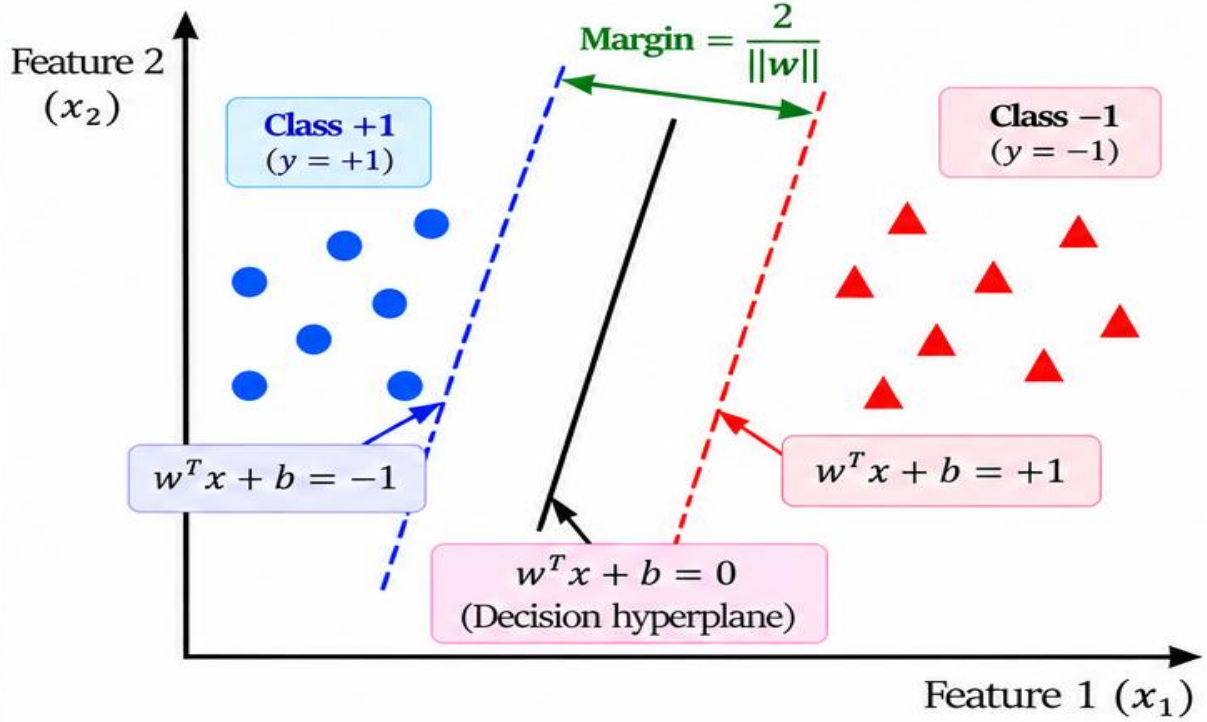
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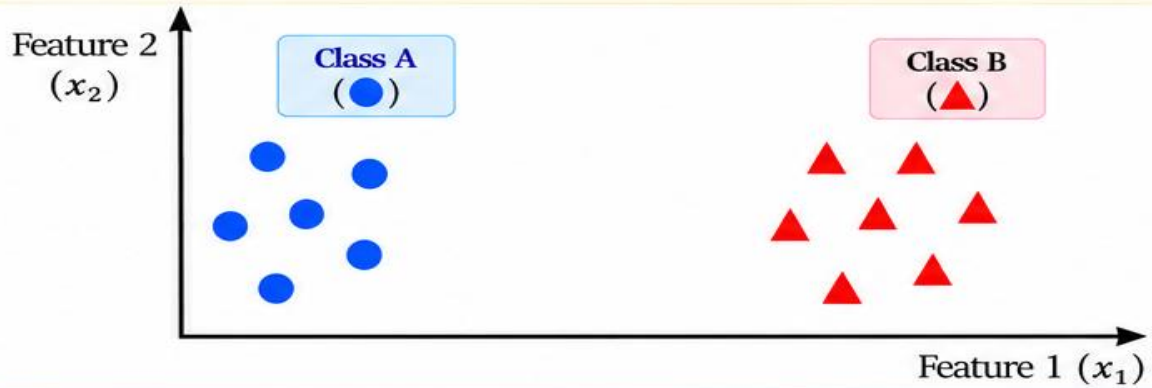
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- Points that lie exactly on these boundaries are the **support vectors**.
- The constraint helps to correctly classify the training data with the maximum possible margin.

Example: Maximum Margin Hyperplane

1 Two classes of data

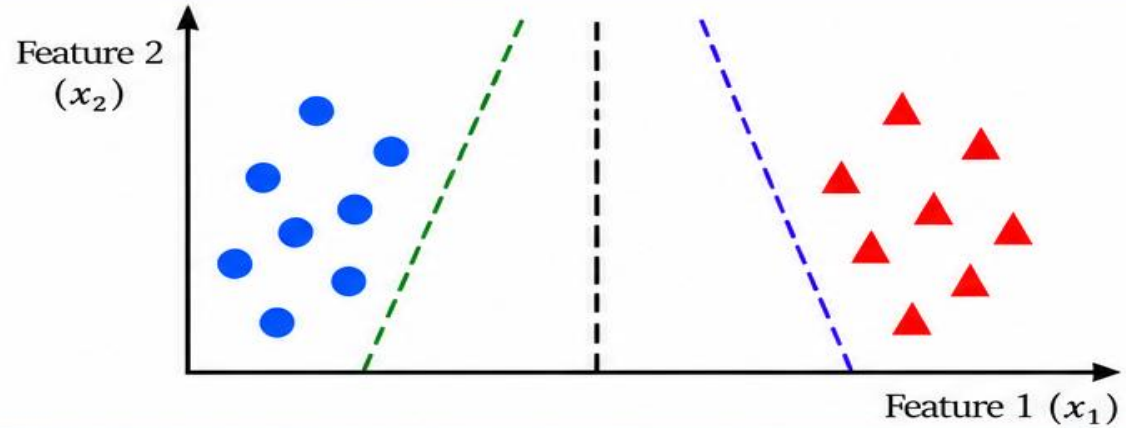
Suppose two classes of data are:

Class A → ● (blue circles)
Class B → ▲ (red triangles)



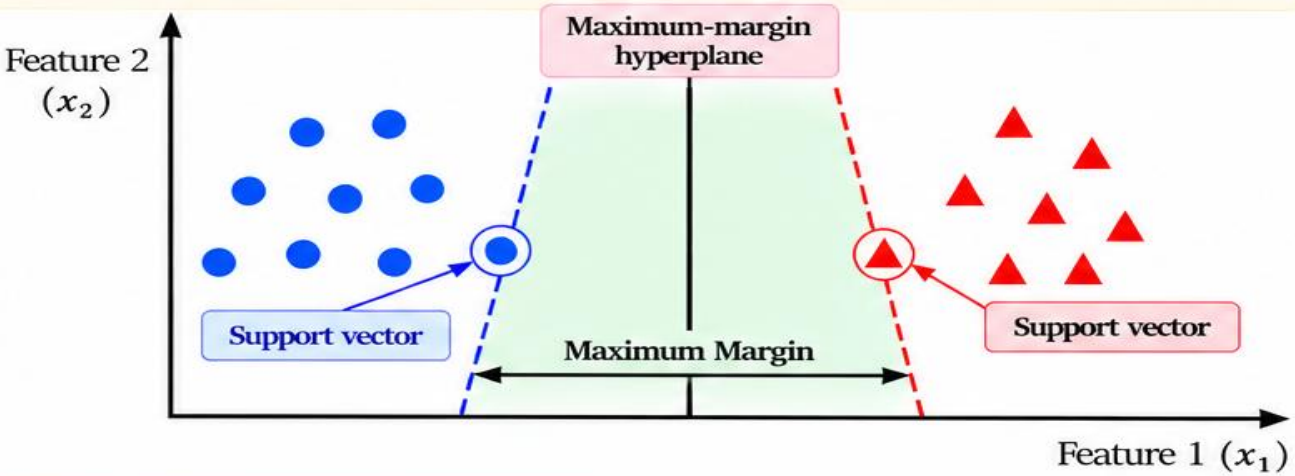
2 Several separating lines are possible

There can be several hyperplanes (decision boundaries) that separate the two classes.



3 SVM chooses the maximum margin hyperplane

SVM selects the hyperplane that provides the largest distance (margin) from the closest points of both classes.



Result

The central line is the maximum-margin hyperplane, which has the largest distance from the closest points of both classes (support vectors).

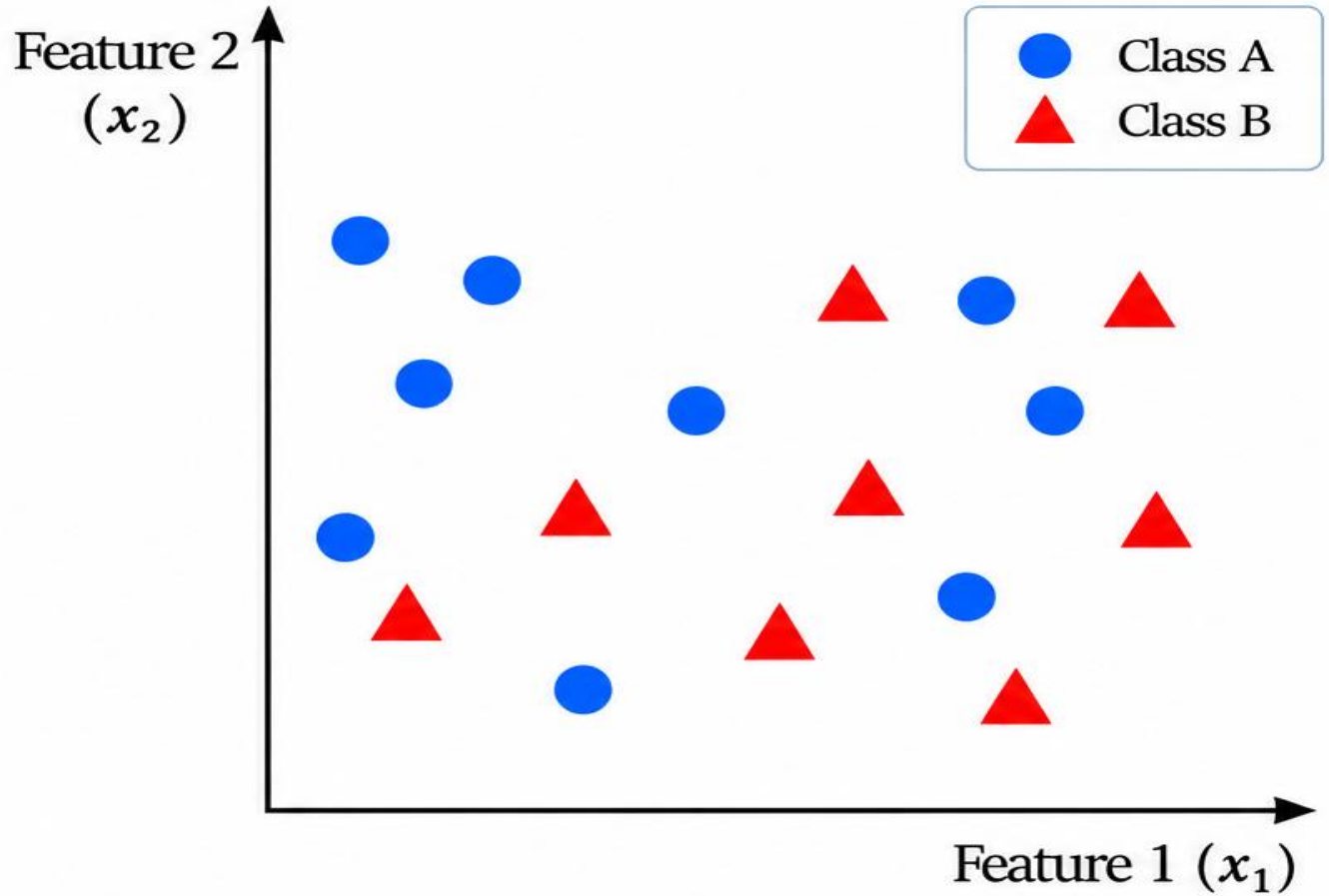
- Maximum-margin hyperplane ($w^T x + b = 0$)
- Negative margin boundary ($w^T x + b = -1$)
- Positive margin boundary ($w^T x + b = +1$)
- Class A
- Class B
- Support vector (closest point to hyperplane)

Non-Linearly Separable Data

- The previous discussion considered **linearly separable data**, where a single straight line or hyperplane can separate the classes.
- However, in real-world datasets, the classes may **overlap**, making it impossible to separate them perfectly using a straight line.

Example

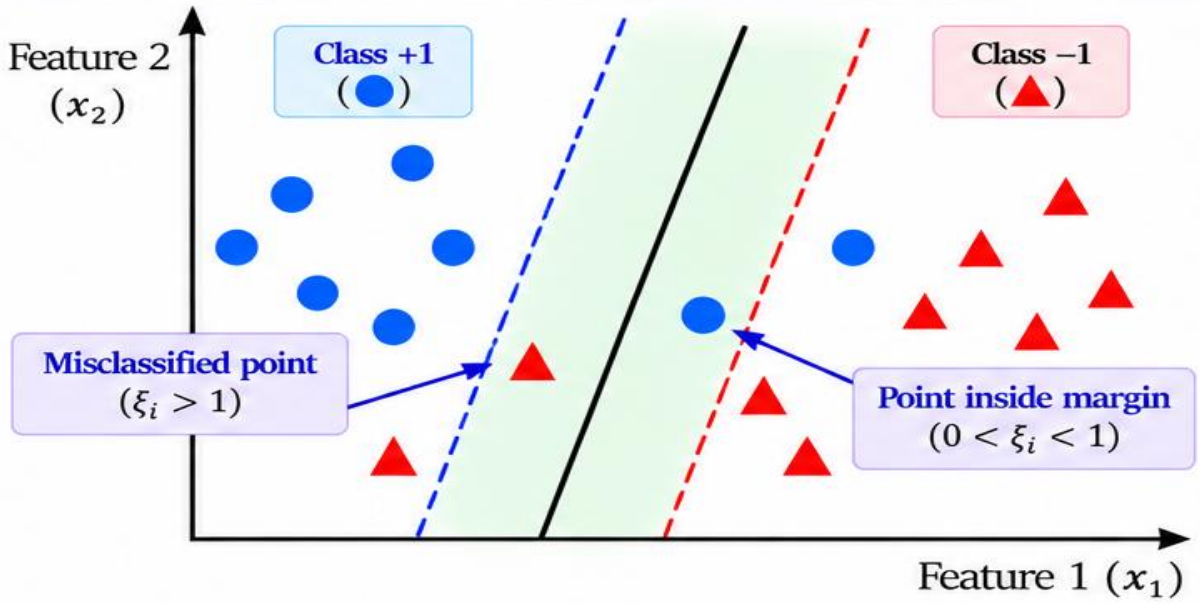
In the following figure, the two classes of data are mixed and cannot be separated by a single straight line.



A single straight line cannot completely separate the two classes.

1 Soft Margin SVM

- When data is not perfectly separable, SVM can allow some data points to:
  - Fall inside the margin
  - Be on the wrong side of the decision boundary
- This approach is called **Soft Margin SVM**.



The objective is to find a balance between
a large margin and **a small number of classification errors**.

Decision hyperplane ($w^T x + b = 0$)

Negative margin boundary ($w^T x + b = -1$)

Positive margin boundary ($w^T x + b = +1$)

2 Slack Variable

- SVM introduces a **slack variable** ξ_i to allow some violations of the margin constraints.
- The constraint becomes:

$$y_i(w^T x_i + b) \geq 1 - \xi_i$$

where:

$$\xi_i \geq 0$$

Meaning of ξ_i

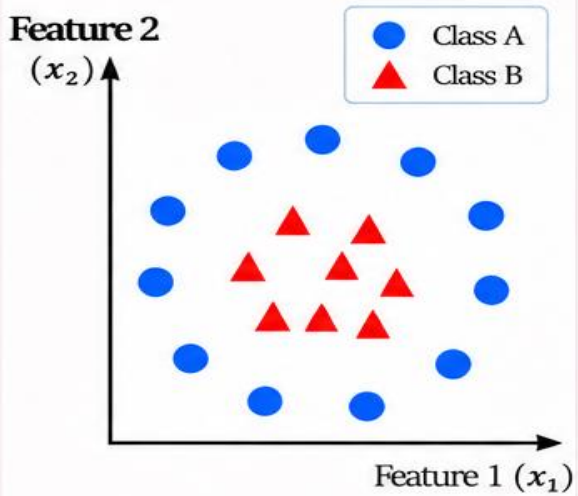
Value of ξ_i	Meaning	Example Position in Plot
$\xi_i = 0$	Correctly classified and outside/on the margin	On or outside margin ($\xi_i = 0$)
$0 < \xi_i < 1$	Correctly classified but inside the margin	Inside margin ($0 < \xi_i < 1$)
$\xi_i > 1$	Misclassified (on the wrong side of the decision boundary)	Misclassified ($\xi_i > 1$)

Using Kernels for Non-Linear Spaces

1 Need for Kernels

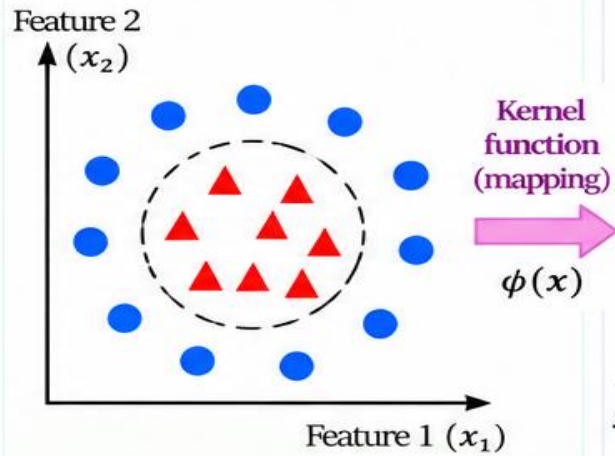
- SVM works very well when the classes can be separated by a straight line or hyperplane.
- However, some datasets are **non-linearly separable**, where a straight line cannot separate the classes.
- In such cases, SVM can use **kernel functions** to handle the non-linear relationship between the data points.

2 Example: Non-Linear Data



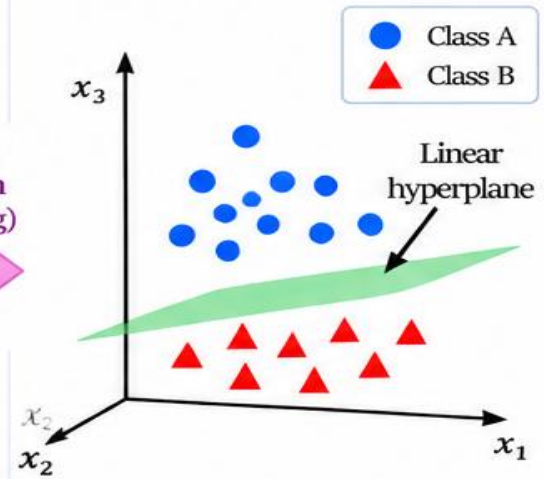
A straight line cannot separate the two classes.

3 Kernel Transformation



Non-linear decision boundary in **original** space

4 Higher-Dimensional Feature Space



After mapping to a higher-dimensional space, the classes become linearly separable.

5 Simple Definition

A kernel function allows SVM to handle **non-linearly separable data** by representing the data in a **higher-dimensional feature space** where the classes may become linearly separable.

Basic Idea of Kernel Method

1 Basic Idea

Non-linear data

↓

Kernel function ($\phi(x)$)

↓

Higher-dimensional feature space

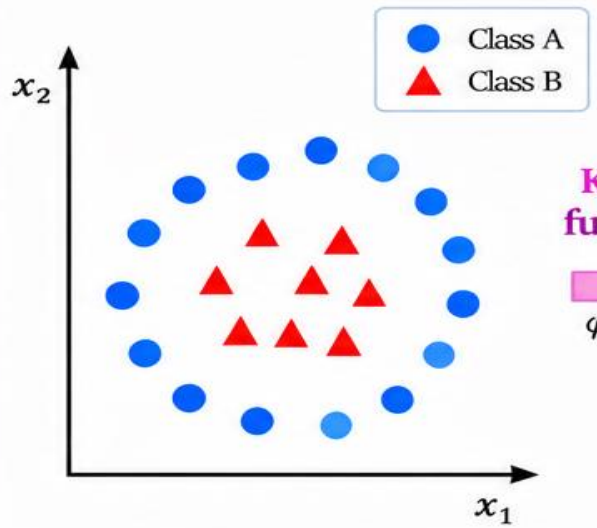
↓

Linear hyperplane

↓

Classification

2 Original Space (Non-linear Data)

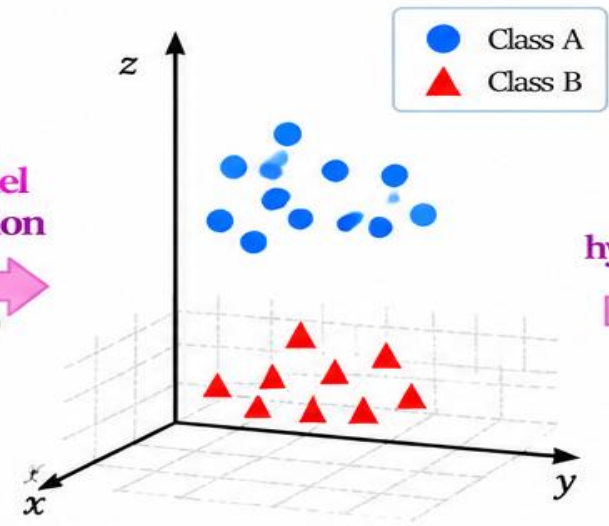


The classes are not linearly separable in the original space. A straight line cannot separate them.

Kernel function

$\phi(x)$

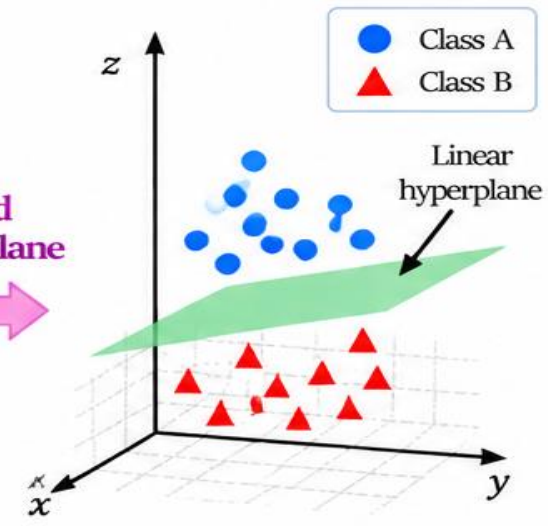
3 Higher-Dimensional Feature Space



Using a kernel function, the data is mapped to a higher-dimensional feature space where the classes become linearly separable.

Find hyperplane

4 Linear Hyperplane in Feature Space



A linear hyperplane can separate the classes in the higher-dimensional feature space.

5 Key Point

Instead of trying to draw a complicated curved boundary directly in the original space,
SVM transforms the data into another feature space using a **kernel function**,
where a **linear hyperplane** can be found for classification.

Mapping to a Higher-Dimensional Space

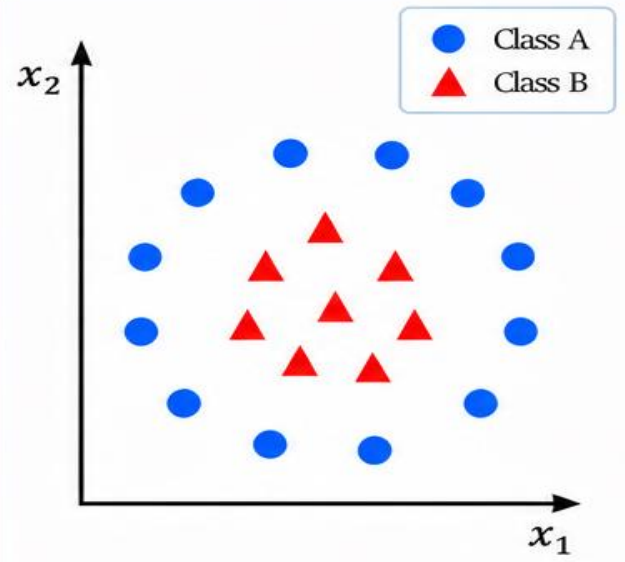
- Consider data with two features: x_1, x_2.
- The data may not be separable in the original two-dimensional space.
- A transformation can map the data to a higher-dimensional space: $\phi(x)$.
- For example:

$$\phi(x_1, x_2) = (x_1, x_2, x_1^2 + x_2^2)$$

- The additional feature can make the classes easier to separate.

Conceptual View

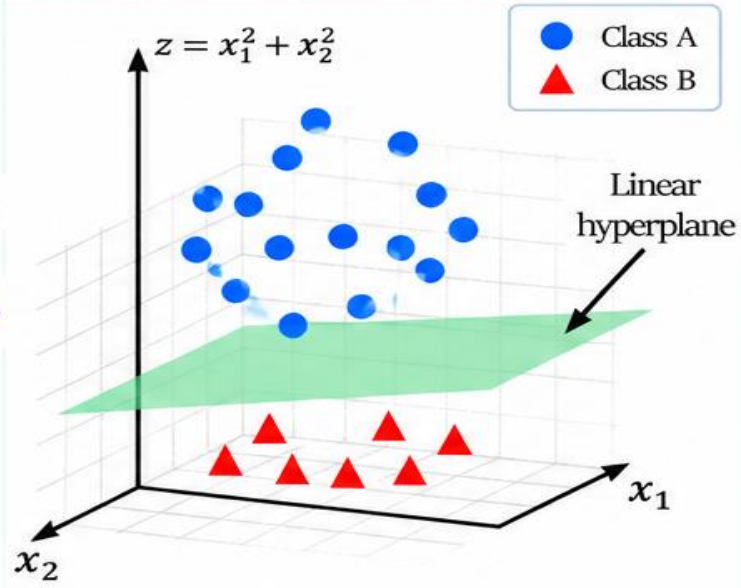
Original Space (2D)



The classes are not linearly separable in the original space. A straight line cannot separate them.

Mapping $\phi(x)$

Higher-Dimensional Space (3D)



In the higher-dimensional space, the classes become linearly separable by a hyperplane.

Key Point

A non-linearly separable dataset in the original 2D space can be mapped to a **higher-dimensional feature space** using a transformation $\phi(x)$, where the classes may become **linearly separable**.

Kernel Trick

1 Explicit Transformation

- Explicitly transforming every data point into a higher-dimensional space can be computationally expensive.
- For example, if we map $\phi(x_1, x_2) = (x_1, x_2, x_1^2 + x_2^2)$ for every data point, we need to compute and store all transformed features.

This can be inefficient when the data is large or the transformed space has many dimensions.

2 Kernel Trick Idea

- Instead of explicitly calculating $\phi(x)$, SVM uses a kernel function to directly compute the similarity between two data points in the transformed space.
- The kernel function is defined as:

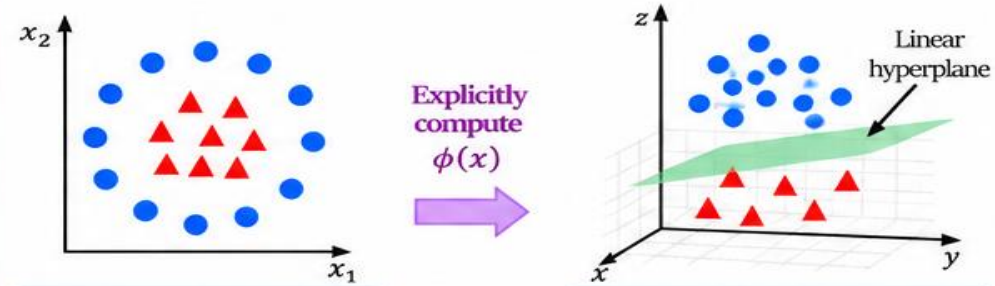
$$K(x_i, x_j) = \phi(x_i)^T \phi(x_j)$$

Simple Meaning

The kernel trick allows SVM to work in a **higher-dimensional feature space** without explicitly calculating all the transformed features.

3 Illustration of Kernel Trick

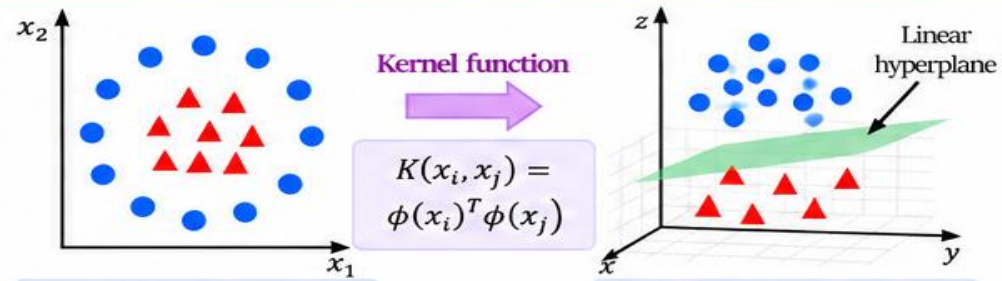
Explicit Transformation (Computationally Expensive)



Original 2D space (Non-linearly separable)

Higher-dimensional space (after explicit transformation)

Using Kernel Trick (No Explicit Transformation)



Original 2D space (Non-linearly separable)

Higher-dimensional space (implicitly using kernel trick)

Key Point

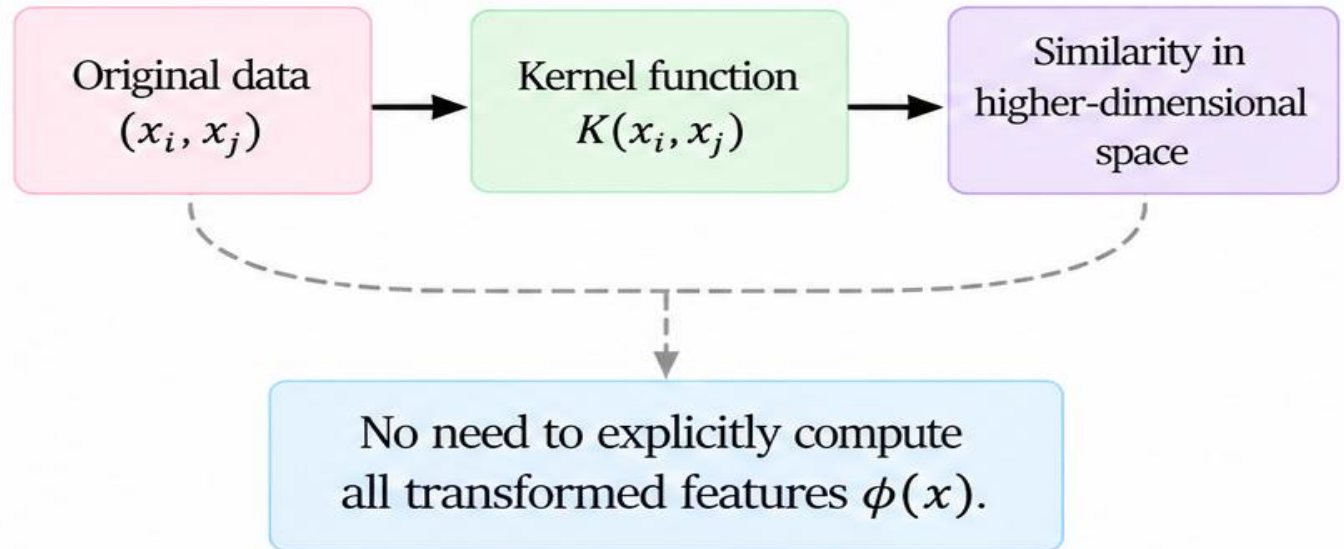
The kernel trick allows **SVM** to work in a **higher-dimensional feature space without explicitly calculating** all the transformed features.

Kernel Trick

- Transforming every data point to a higher-dimensional space can be **computationally expensive**.
- SVM uses the **kernel trick** to avoid explicitly calculating the transformed features.
- Instead of computing $\phi(x)$ for each data point, the kernel function **directly computes the similarity** between two data points in the transformed space.

$$K(x_i, x_j) = \phi(x_i)^T \phi(x_j)$$

Idea of Kernel Trick



Simple Meaning

The kernel trick allows **SVM** to work in a **higher-dimensional feature space without explicitly calculating** all the transformed features.

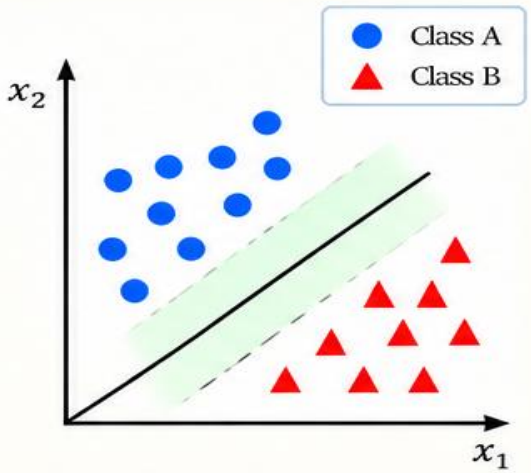
Common Kernel Functions

1 Linear Kernel

Used when the data is **approximately** linearly separable.

$$K(x_i, x_j) = x_i^T x_j$$

It does not perform a non-linear transformation.



Suitable for linearly separable or simple data.

2 Polynomial Kernel

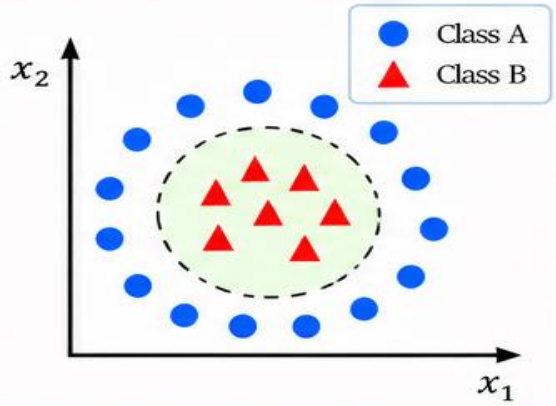
Can model non-linear relationships.

$$K(x_i, x_j) = (\gamma x_i^T x_j + r)^d$$

where,

γ – scaling parameter
r – constant
d – degree of the polynomial

Example: $d = 2$ gives a quadratic relationship.



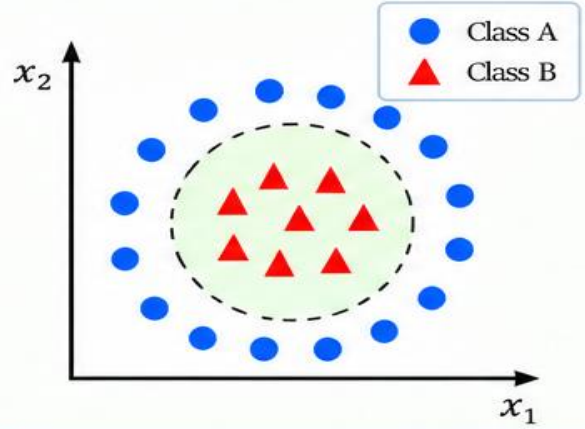
Useful for non-linear patterns (polynomial relationships).

3 Radial Basis Function (RBF) Kernel

Widely used for non-linear classification.

$$K(x_i, x_j) = \exp\left(-\gamma \|x_i - x_j\|^2\right)$$

It measures similarity based on the distance between data points. Points that are closer together have a higher kernel similarity.



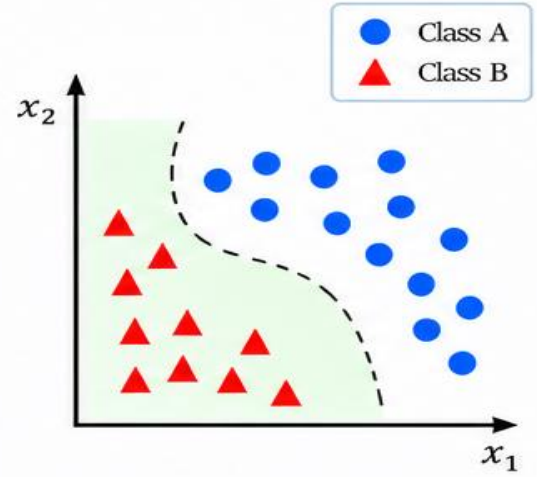
Effective for complex non-linear data patterns.

4 Sigmoid Kernel

Based on the sigmoid function.

$$K(x_i, x_j) = \tanh(\gamma x_i^T x_j + r)$$

It can be useful for certain non-linear relationships.



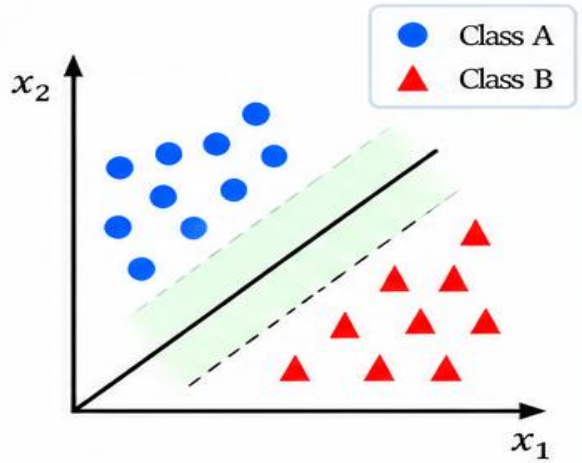
Useful for certain non-linear patterns.

Comparison of Common Kernels

Kernel	Main Use	Nature
Linear	Simple, linearly separable data	Linear
Polynomial	Polynomial relationships	Non-linear
RBF (Gaussian)	Complex non-linear patterns	Non-linear
Sigmoid	Certain neural-network-like relationships	Non-linear

Linear Kernel

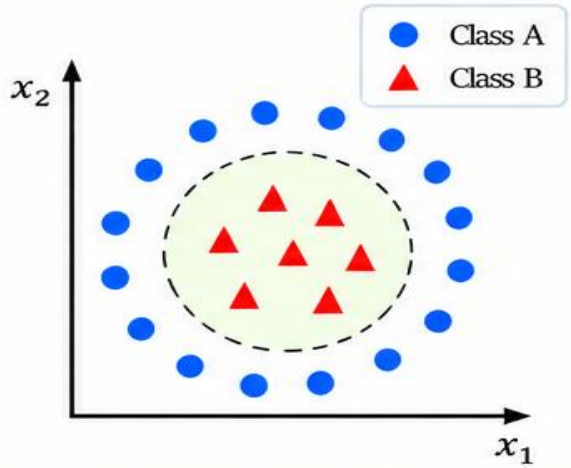
Used for simple, linearly separable data.



Gives a straight boundary.

Polynomial Kernel

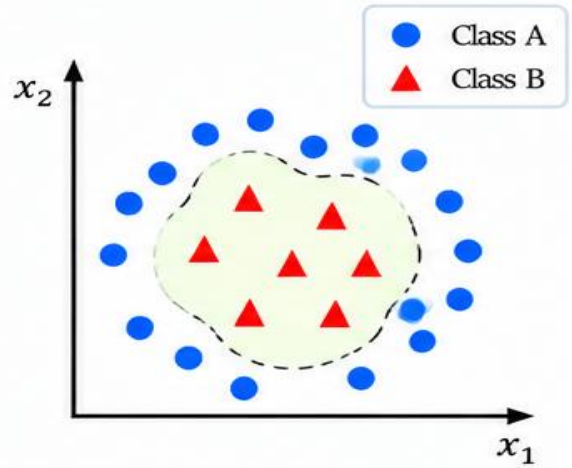
Can model non-linear (polynomial) relationships.



Gives a curved boundary (polynomial shape).

RBF Kernel

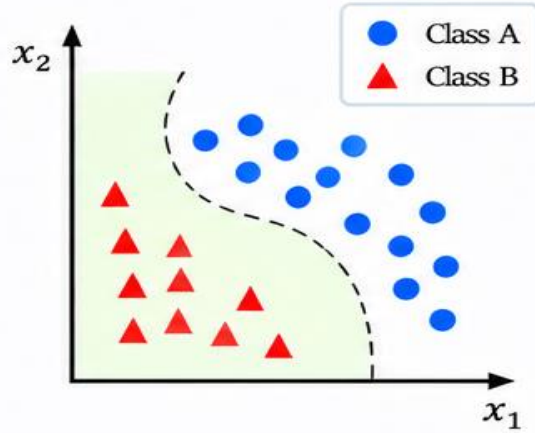
Useful for complex non-linear patterns.



Gives a flexible curved boundary for complex patterns.

Sigmoid Kernel

Useful for certain neural-network-like relationships.

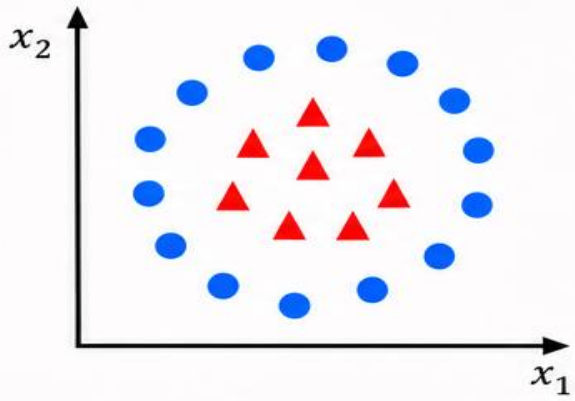


Gives a sigmoid-like non-linear boundary.

Advantages of Kernel Methods

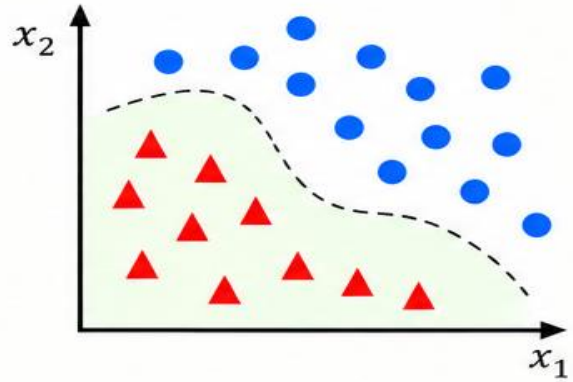
1 Handle non-linearly separable data

Can be used for data that cannot be separated by a straight line.



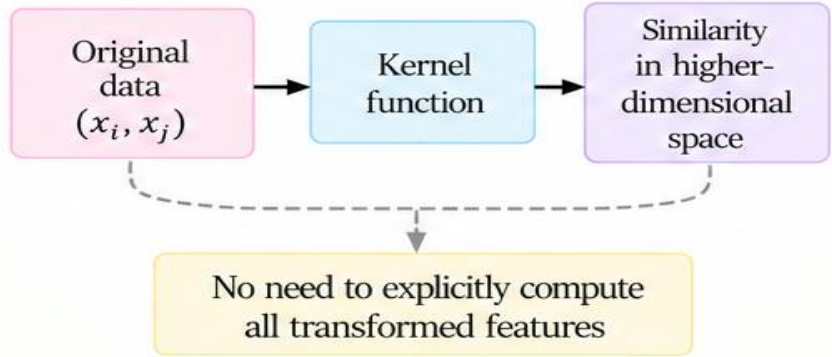
2 Allow complex decision boundaries

Helps SVM create flexible and complex decision boundaries.



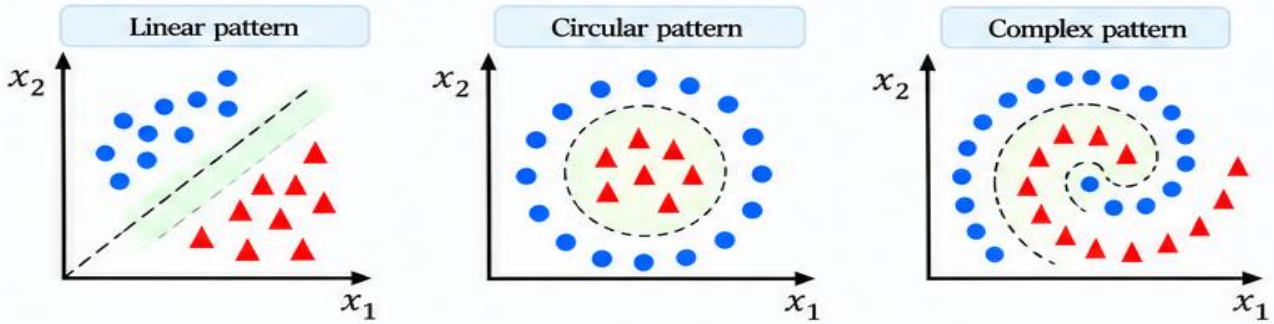
3 Avoid explicit high-dimensional computation

Uses the kernel trick to work in a higher-dimensional space without explicitly computing all transformed features.



4 Work with different types of data patterns

Can handle various data distributions such as linear, circular, or more complex shapes.



5 RBF and polynomial kernels can capture non-linear relationships

RBF and polynomial kernels are effective in modelling complex non-linear patterns in the data.

